

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3529496303641594, median 0.3134642911278599, std: 0.2112861744534744
Reprojection error (cam1): mean 0.29877728409501575, median 0.2697557052583361, std: 0.17420998825637365
Gyroscope error (imu0): mean 0.13616467668481824, median 0.09161827181722251, std: 0.13272606483411747
Accelerometer error (imu0): mean 0.5564267093259277, median 0.4535891463699594, std: 0.5850162704044976

Residuals

Reprojection error (cam0) [px]: mean 0.3529496303641594, median 0.3134642911278599, std: 0.2112861744534744
Reprojection error (cam1) [px]: mean 0.29877728409501575, median 0.2697557052583361, std: 0.17420998825637365
Gyroscope error (imu0) [rad/s]: mean 0.07974413409058377, median 0.05365576396770994, std: 0.07773032896003243
Accelerometer error (imu0) [m/s^2]: mean 0.047391107870358405, median 0.03863238734618536, std: 0.04982609337758189

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ 0.99991552 -0.00960214 -0.00876055 0.07430058]
 [ 0.00966822 0.99992489 0.00753257 -0.00147473]
 [ 0.00868756 -0.00761663 0.99993325 -0.00757521]
 [ 0.         0.         0.         1.        ]]
```

T_ic: (cam0 to imu0):

```
[[ 0.99991552 0.00966822 0.00868756 -0.07421423]
 [-0.00960214 0.99992489 -0.00761663 0.00213037]
 [-0.00876055 0.00753257 0.99993325 0.00823673]
 [ 0.         0.         0.         1.        ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
0.002561957316995907

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.99988863 -0.00917143 -0.01177344 0.00014254]
[0.00924868 0.99993595 0.00652403 -0.00169414]
[0.01171285 -0.00663219 0.99990941 -0.00806322]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.99988863 0.00924868 0.01171285 -0.00003241]
[-0.00917143 0.99993595 -0.00663219 0.00164186]
[-0.01177344 0.00652403 0.99990941 0.00807522]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.0025758884478283106

Baselines:

Baseline (cam0 to cam1):
[[0.99999537 0.00040772 -0.0030162 -0.07417994]
[-0.00041077 0.9999994 -0.0010122 -0.00019656]
[0.00301579 0.00101344 0.99999494 -0.00071063]
[0. 0. 0. 1.]]

baseline norm: 0.07418360667264533 [m]

Gravity vector in target coords: [m/s^2]
[9.80533585 -0.15072084 -0.03309128]

Calibration configuration

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cam0

Camera model: pinhole
Focal length: [401.608628861003, 399.428391132301]
Principal point: [319.7343936841441, 190.34744354361882]
Distortion model: radtan
Distortion coefficients: [-0.0076233519570338935, -0.0026557069530887645, -0.0008795023429080058, -0.00016680927611684287]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.03432 [m]
Spacing 0.010296000000000001 [m]

cam1

Camera model: pinhole
Focal length: [402.27012127979714, 400.2659622399602]
Principal point: [320.98190171333647, 190.80434036162683]
Distortion model: radtan
Distortion coefficients: [-0.0127486230660343, 0.009669430813801899, -0.0005299371877853309, 0.00021110128372654215]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.03432 [m]
Spacing 0.010296000000000001 [m]

IMU configuration

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IMU0:

Model: scale-misalignment
Update rate: 200

Accelerometer:

Noise density: 0.006022459594664195

Noise density (discrete): 0.08517044037618077

Random walk: 7.117282780349031e-06

Gyroscope:

Noise density: 0.04141134055333757

Noise density (discrete): 0.5856447944658094

Random walk: 0.006936521501009744

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

[0. 1. 0. 0.]

[0. 0. 1. 0.]

[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

Gyroscope:

M:

[[0.98451906 0. 0.]

[0.00690073 0.98432363 0.]

[-0.00331977 -0.01625885 0.9962188]]

A [(rad/s)/(m/s²)]:

[[0.00008826 0.00001289 0.00124125]

[0.0003909 0.00038689 -0.00001042]

[-0.00030352 -0.00046336 -0.00018719]]

C_gyro_i:

[[0.99998824 0.00181777 -0.00449566]

[-0.00186435 0.99994441 -0.0103783]

[0.00447654 0.01038656 0.99993604]]

Accelerometer:

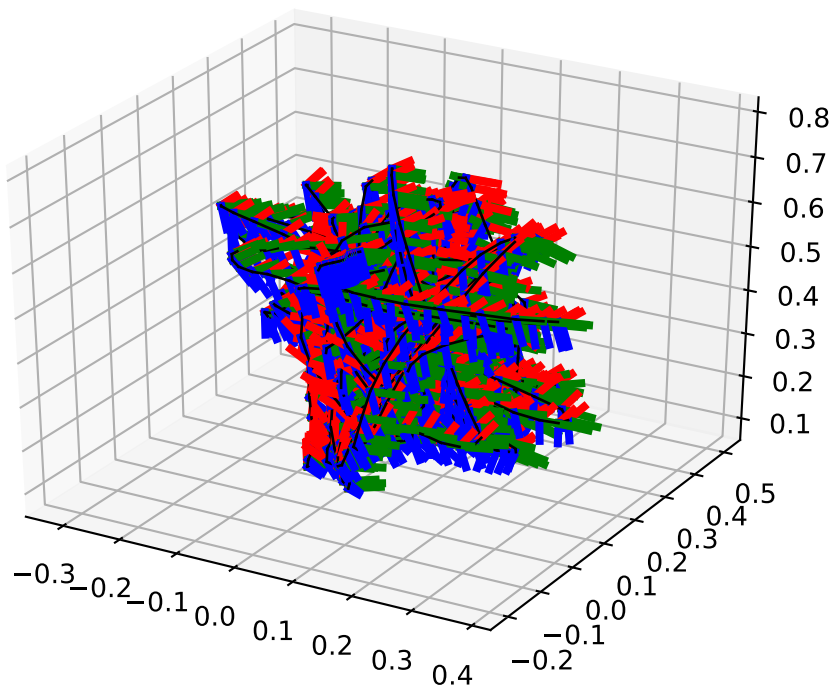
M:

[[0.99825897 0. 0.]

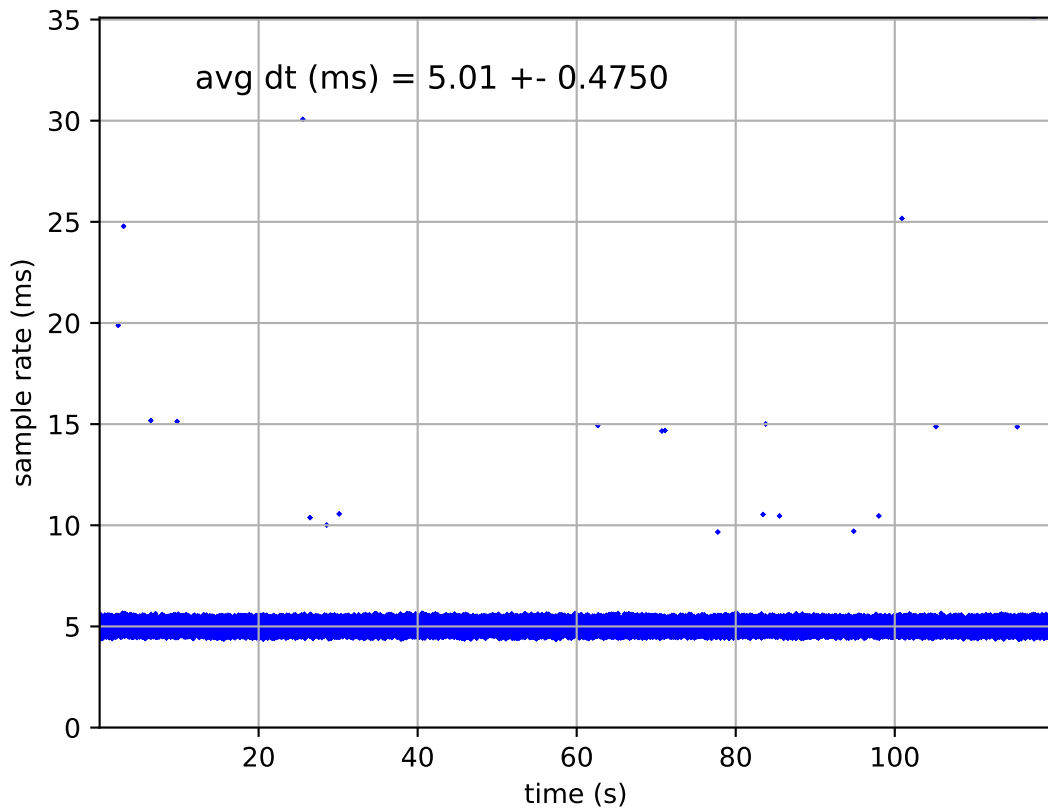
[-0.0213733 0.99693525 0.]

[0.03077868 0.00171986 1.0012464]]

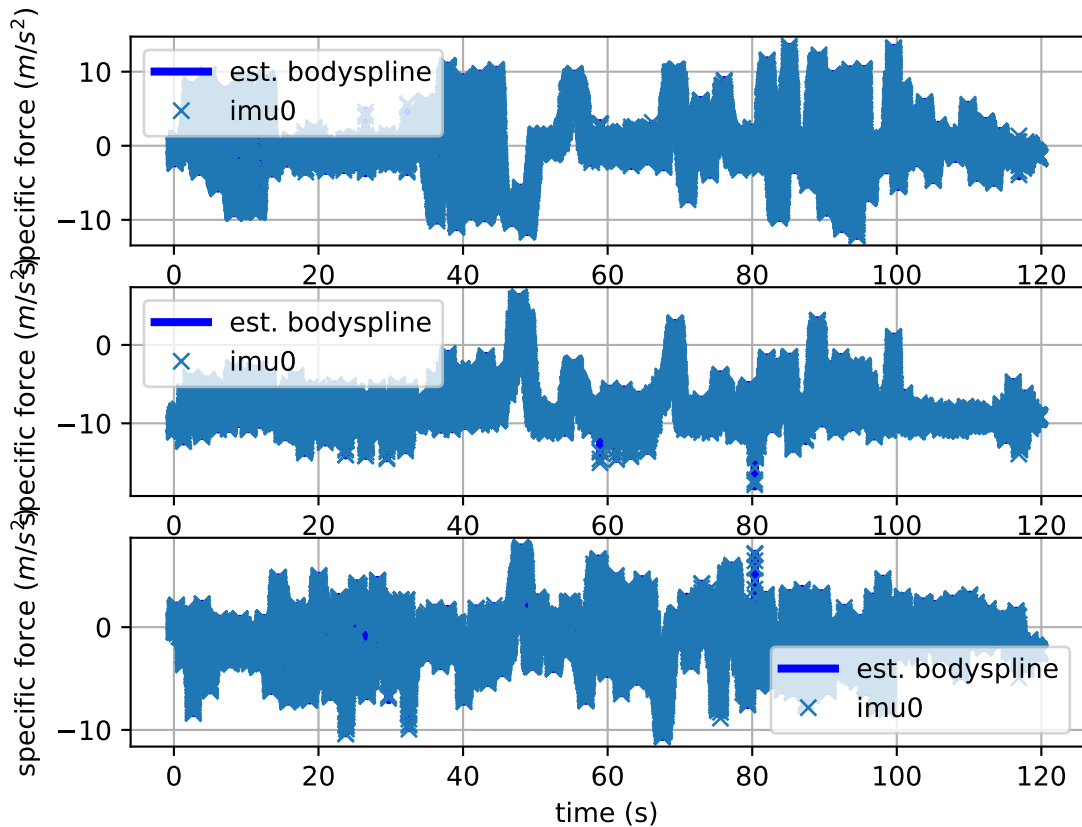
imu0: estimated poses



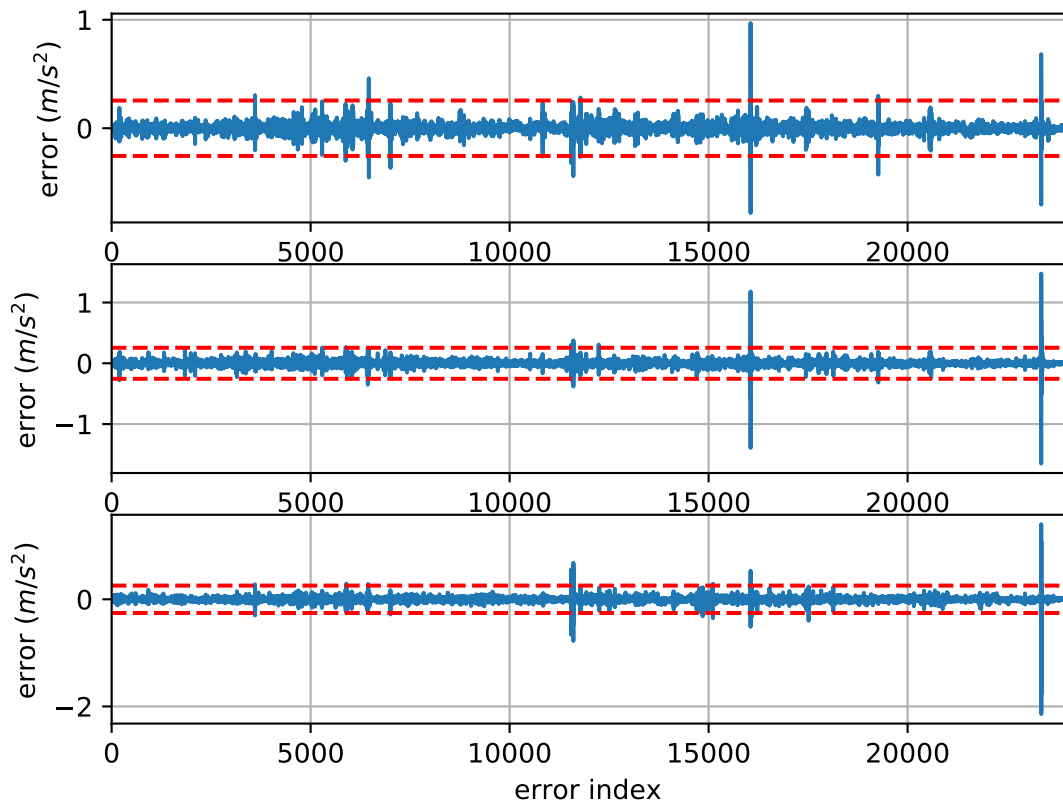
imu0: sample inertial rate



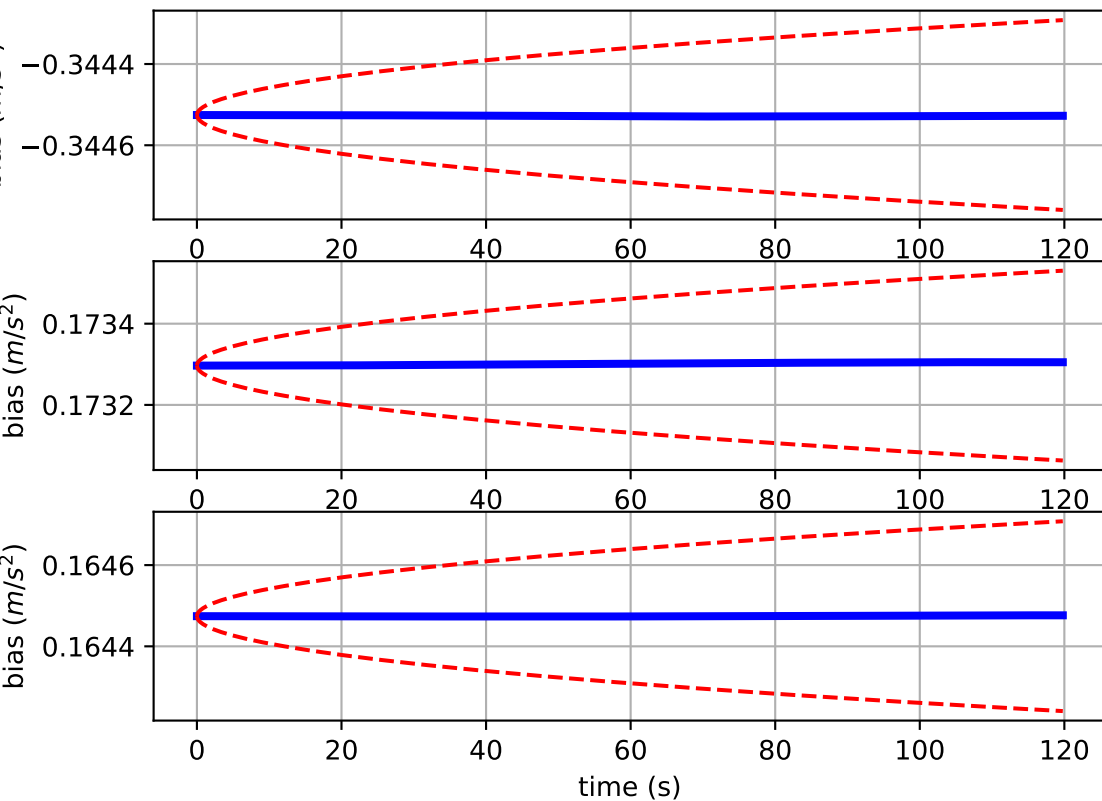
Comparison of predicted and measured specific force (imu0 frame)



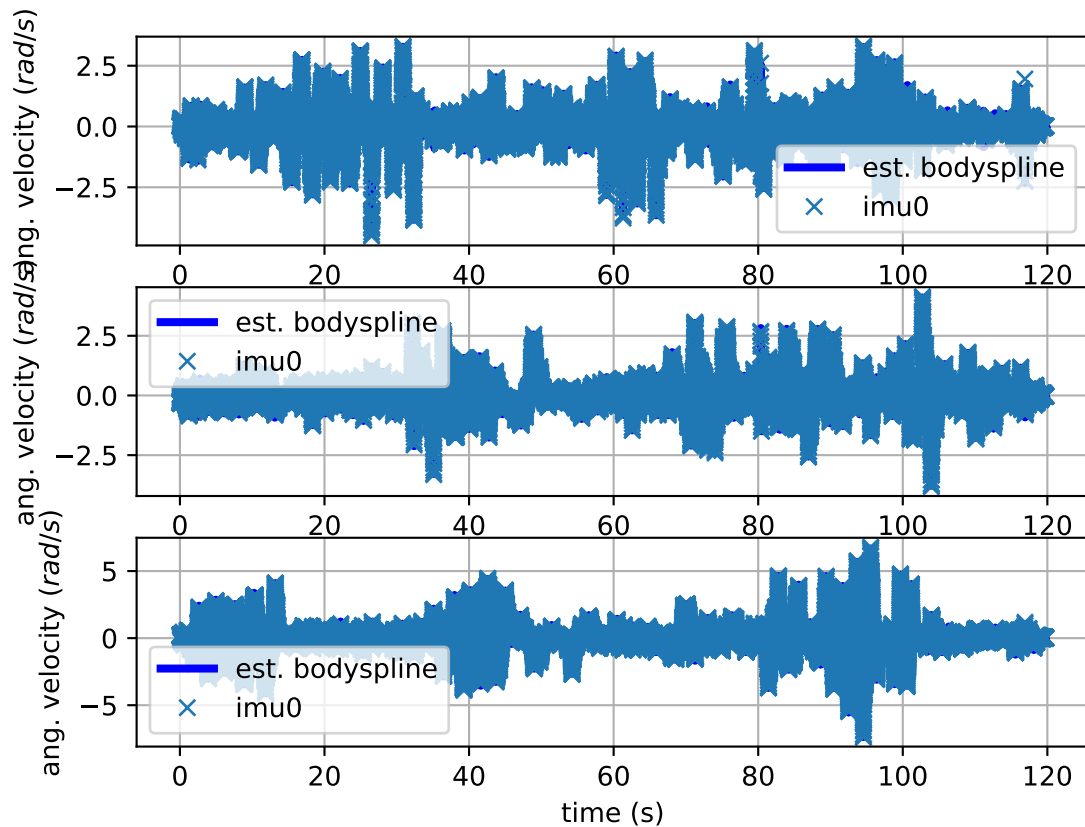
imu0: acceleration error



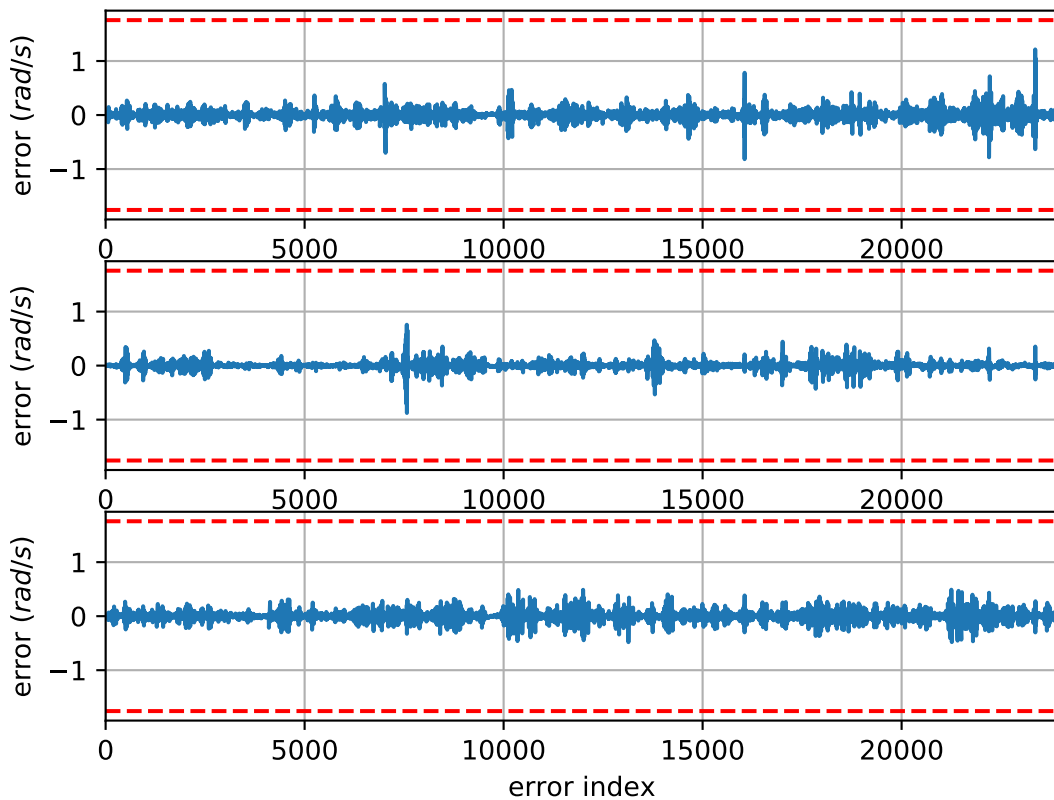
imu0: estimated accelerometer bias (imu frame)



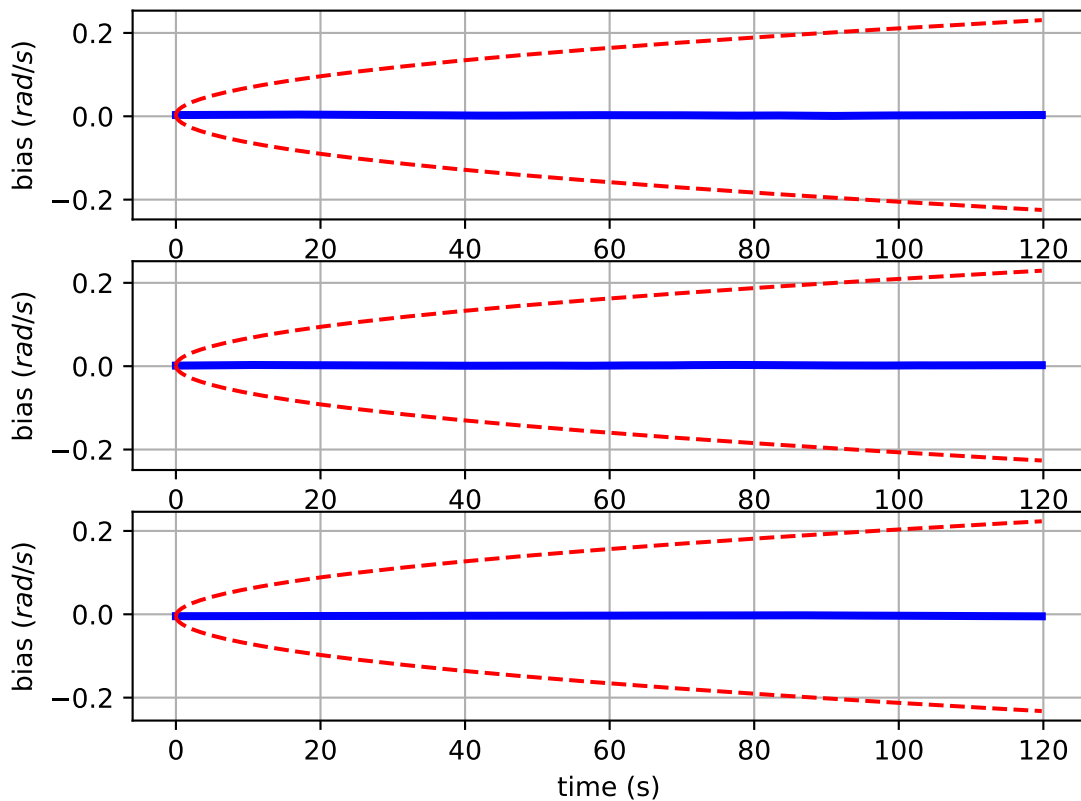
Comparison of predicted and measured angular velocities (body frame)



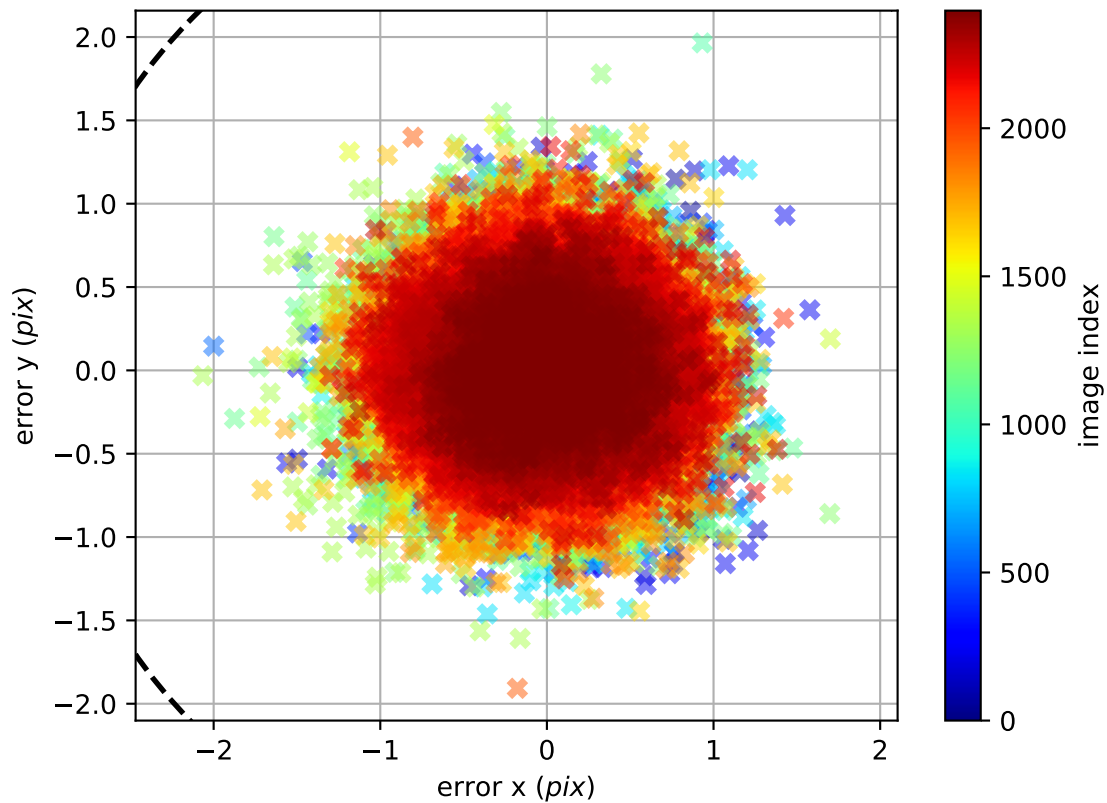
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

